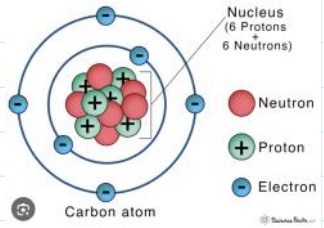
Lecture 1: Introduction to Electrical Engineering

Electrical Engineering is science of electrons.

Materials → Molecules → Atoms → Particles
- Particles → Neutrons, Protons } Nucleus
- Particles → electrons } Shell



Atom's mass number = Sum of neutrons and protons
= No of neutrons + No of protons

Atomic number = Number of protons
= Number of electrons

Particle Details:

Particle	symbol	Mass	charge	Location.
electron	e^-	9.1094×10^{-31} kg	-1.6×10^{-19} C	Outside the nucleus
proton	p^+	1.6726×10^{-27} Kg	1.6×10^{-19} C	Inside nucleus
neutron	n^0	1.6726×10^{-27} kg	0	Inside nucleus

- When electrons are removed from the item, it becomes positively charged
- When electrons are excess in the atom it becomes negatively charged

The nature of material decide the number and distribution of electrons in the electron cloud space.

Materials → Conductors, Semiconductors, Insulators

Conductors: In some materials, the electrons are relatively free of the nucleus, and allows flow of currents. These materials are called conductors. Examples: metals, iron, copper, silver, gold etc.

Semiconductors: Relatively electrons are few to move. Their movement can be controlled by applying the electric field. These

Semiconductors : Relatively electrons are ... weak interactions can be controlled by applying the electric field. These materials such as Silicon, Germanium are basis of electronic science. These find application in electronics, diodes, transistors, digital processors. RAM, ROM memories etc

Insulators : Insulators have no free electrons to move, hence electric current does not flow freely. Thus, it has a high resistivity. materials such as, wood, glass paper, polymers, rubber, mica are insulators. Insulators are normally used for insulators.

Charge, Current, Voltage, Power, and Energy

Electron : e

In electrical systems, Charge and energy are fundamental quantities.

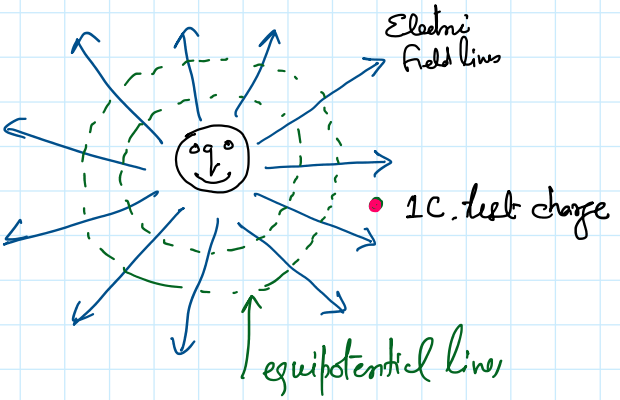
$$e \downarrow$$

$$q = ne \text{ (Coulomb)}$$

$$\downarrow$$

$$i = \frac{dq}{dt} \text{ (Ampere)}$$

$$1 \text{ Ampere} = \frac{1 \text{ Coulomb}}{\text{Sec}}$$



A charge forms an Electric field ($\vec{E}$).
In presence of this electric field, if an unit test charge of 1 Coulomb is brought from ∞ to point P, then the work done to do this.

Gauss's Law: Charge enclosed = closed surface integration of electric flux density

$$q = \oint D_\perp \cdot ds$$

$$q = \epsilon_0 E_r 4\pi r^2$$

$$\Rightarrow E_r = \frac{q}{4\pi\epsilon_0} \cdot \frac{1}{r^2}$$

Electric field Intensity

$$\Rightarrow E_r = \frac{q}{4\pi\epsilon_0} \cdot \frac{1}{r^2}$$

$$\boxed{\vec{E}_r = \frac{q}{4\pi\epsilon_0} \cdot \frac{1}{r^2} \hat{a}_r}$$

$\epsilon = \epsilon_0 * \epsilon_r$ (ϵ: air permittivity)

Voltage

$$V = -\int_{\infty}^{r} \vec{E}_r \cdot d\vec{r} \qquad \vec{E}_r = \frac{1}{4\pi\epsilon} \cdot \frac{q}{r^2} \hat{a}_r$$

$$V(r) = -\frac{1}{4\pi\epsilon} \cdot \int_{\infty}^{r} \frac{q}{r^2} dr = \frac{q}{4\pi\epsilon r}$$

Force and Work Done

Coulomb's Law

The force between q and q_{test} is

$$\vec{F} = \frac{1}{4\pi\epsilon} \cdot \frac{q \cdot q_{test}}{r^2} \hat{a}_r$$

$W = \int \vec{F} \cdot d\vec{r}$, $\int F(r)\, dr$, $W = \int \vec{F} \cdot d\vec{s}$

$$W = \vec{F} \cdot \vec{r} = \frac{1}{4\pi\epsilon} \cdot \frac{q\, q_{test}}{r^2} \cdot r = \frac{1}{4\pi\epsilon} \cdot \frac{q \cdot q_{test}}{r}$$

or $$\boxed{\frac{W}{q_{test}} = \frac{1}{4\pi\epsilon} \cdot \frac{q}{r} = V}$$

Thus $$\boxed{W = V q_{test} = V q} \quad \text{(Joules)}$$

$q \rightarrow$ Electrostatic field $\rightarrow$ Electro-motive force (EMF), V

$i \rightarrow$ Magnetic field $\rightarrow$ Magneto-motive force (MMF), NI

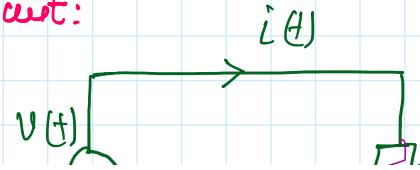
Power (P):

$$p = \frac{dw}{dt} = \frac{d(V \cdot q)}{dt} = V \cdot \frac{dq}{dt} = V i \quad \text{(VA)}$$

Power = Voltage * Current (Watts)

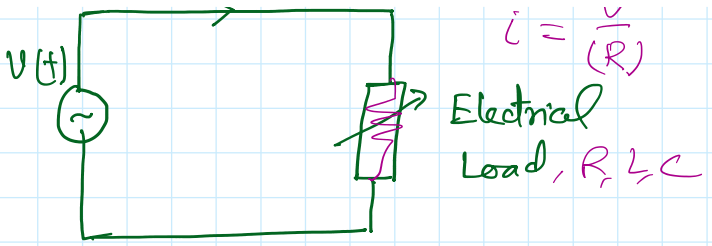
= Cause * Effect

Consider the following circuit:



$i \propto V$ $\quad V = IR$

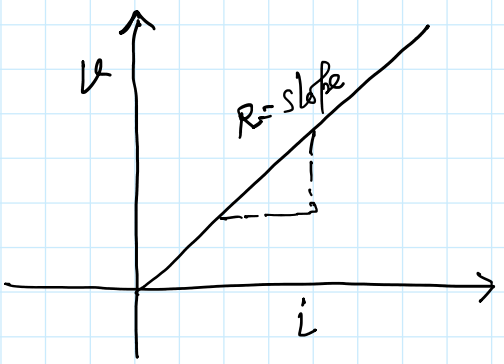
$i = \frac{V}{(R)}$

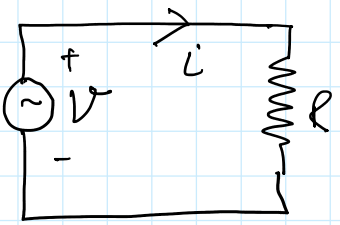


Electrical Load : Resistance, Inductance, Capacitance or their combinations.

Resistance: $i \propto v$; $i = \frac{V}{R}$ or $V = iR$

where R is proportionality constant, known as resistance. This is known as Ohm's law.

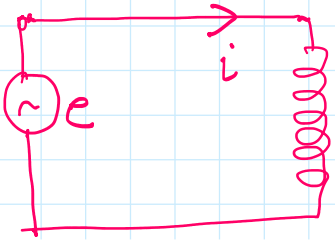


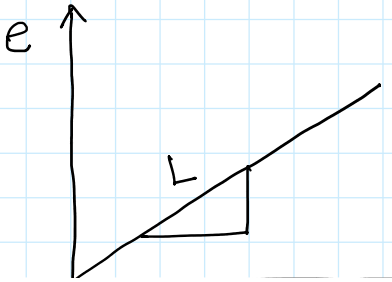


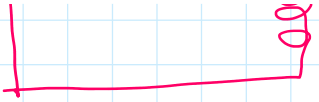
Inductance: is deduced from the Faraday's Law of Induction and Henry's Law.

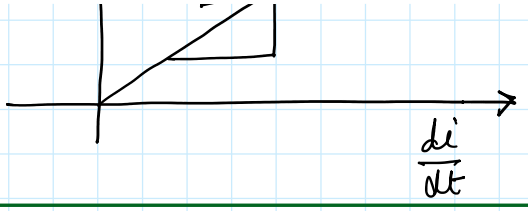
$$e = N \cdot \frac{d\Phi}{dt} = L \frac{di}{dt} \Rightarrow i_L = \frac{1}{L}\int e_L(t)\, dt$$

$$\frac{e}{di/dt} = L$$ = Inductance of circuit.



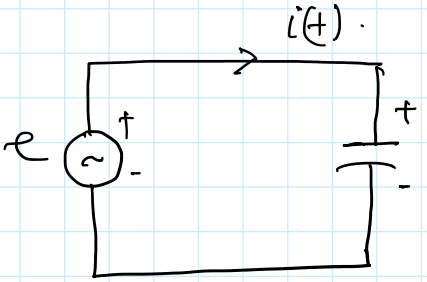






Capacitance: $C = \frac{\text{Charge}}{\text{voltage}}$

$= \frac{\int i_c\,dt}{e}$



or $e = V_C = \frac{1}{C}\int i_c\,dt$

or $i_c = C\frac{dV_C}{dt}$

$\Rightarrow C = \frac{\int i_c\,dt}{e}$

